Common teamwork risks in student software project development

1. Uneven Workload Distribution

- Some team members may end up doing most of the work while others contribute minimally ("free riders").
- Leads to burnout and resentment.

2. Poor Communication

- Misunderstandings due to unclear or infrequent communication.
- Delayed feedback loops cause rework and missed deadlines.

3. Lack of Role Clarity

- Ambiguity over who is responsible for which tasks.
- Overlapping or missed responsibilities create conflicts or gaps.

4. Conflicting Schedules

- Difficulty finding common meeting times due to varying personal schedules.
- Leads to asynchronous collaboration challenges and delays.

5. Inconsistent Skill Levels

- Skill gaps between team members may hinder progress.
- Stronger members may have to train others, slowing down development.

6. Decision-Making Conflicts

- Disagreements on design, technology stack, or project direction.
- Can escalate into interpersonal conflicts if not managed well.

7. Lack of Commitment / Motivation

- Team members may prioritize other courses, part-time jobs, or personal interests.
- Leads to procrastination and last-minute work.

8. Version Control & Code Integration Issues

- Inexperience with Git or collaboration tools may cause merge conflicts.
- Leads to overwriting others' work or corrupting codebase.

9. Scope Creep

- Team members or stakeholders keep adding new features beyond the agreed scope.
- Results in overwork and project incompletion.

10. Ineffective Project Planning

- Poorly defined timelines and unrealistic estimates.
- Tasks are left unfinished close to deadlines.

11. Lack of Leadership / Project Management

- No one taking charge to ensure progress, resolve disputes, or track deliverables.
- Leads to disorganization and missed milestones.

12. Dependency on Key Members

- Over-reliance on a few members for critical tasks.
- Risk increases if these members become unavailable (illness, emergencies).

13. Cultural or Personality Clashes

- Differences in work styles, communication preferences, or cultural expectations.
- Can cause friction and hinder collaboration.

14. Inadequate Testing & QA

- Team might skip systematic testing due to time pressure.
- Results in bugs in the final deliverable, affecting project grading.

15. Poor Documentation

- Failure to maintain proper documentation (code comments, user guides).
- Difficult for other team members to understand or continue work.

Project Teamwork Risks to Mitigation Mapping

Risk ID	Risk Description	Impact	Likelihood	Priority	Mitigation Strategy
R1	Uneven Workload Distribution	High	Medium	High	Define roles early, use task tracking tools
R2	Poor Communication	High	High	High	Set up regular meetings and communication tools (Slack, Teams)
R3	Lack of Role Clarity	Medium	High	High	Document responsibilities in a Team Charter
R4	Conflicting Schedules	Medium	High	Medium	Use shared calendars; agree on fixed weekly syncs
R5	Inconsistent Skill Levels	Medium	Medium	Medium	Pair programming & knowledge-sharing sessions
R6	Decision-Making Conflicts	High	Medium	High	Establish voting mechanism; assign final authority
R7	Lack of Commitment / Motivation	High	Medium	High	Peer evaluations; celebrate milestones
R8	Version Control & Code Integration Issues	High	Medium	High	Mandatory Git training; daily commits
R9	Scope Creep	High	Medium	High	Freeze scope early; maintain change log
R10	Ineffective Project Planning	High	Medium	High	Use agile sprints; define deliverables weekly
R11	Lack of Leadership / Project Management	High	Medium	High	Rotate leadership roles per milestone
R12	Dependency on Key Members	High	Medium	High	Cross-train team members on critical tasks
R13	Cultural or Personality Clashes	Medium	Medium	Medium	Team bonding activities; conflict resolution process
R14	Inadequate Testing & QA	High	Medium	High	Define testing tasks in sprint planning; assign QA roles
R15	Poor Documentation	Medium	High	High	Documentation checkpoints per sprint